

LAB 1

REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #09

FACULTY RESEARCH GRANT & PUBLICATION TRACKER

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1. Complete Requirements Table

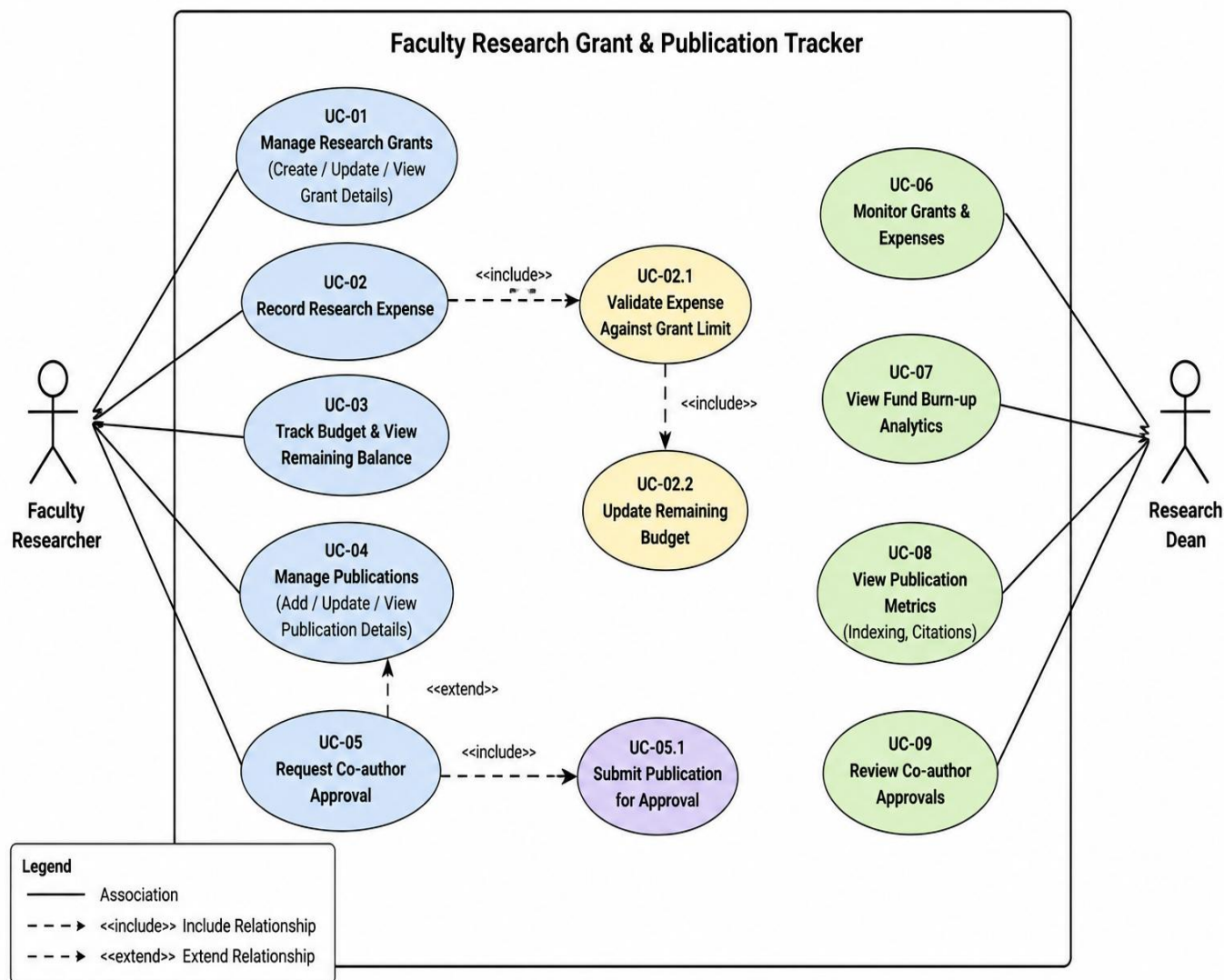
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Faculty Researchers to create and maintain research grant records, including grant allocation and approved funding limits.	High	Pass: A researcher can create/update a grant record and the system displays the correct allocated amount and funding limit. Fail: Grant details are missing, inconsistent, or cannot be updated.	Provides a central record of sponsored research grants and their approved funding.
FR-002	Functional	The system shall allow Faculty Researchers to record equipment procurement and other research expenses, calculate the remaining grant	High	Pass: A valid expense updates the remaining balance accurately; an expense exceeding the available grant	Prevents grant overspending and keeps financial information accurate.

		balance, and block expenses that exceed the grant cap.		amount is rejected. Fail: The system allows an overdraft or calculates the balance incorrectly.	
FR-003	Functional	The system shall allow Faculty Researchers to record publications and maintain publication metrics including indexing status and citation information.	High	Pass: A publication can be recorded and its indexing/citation information can be stored and updated. Fail: Publication records cannot be maintained or metrics are not reflected correctly.	Enables centralized tracking of faculty publication output and impact.
FR-004	Functional	The system shall support a co-author approval workflow for publications and allow the approval status to be tracked.	Medium	Pass: A publication can be submitted for co-author approval and its status changes according to approval actions. Fail: Approval requests cannot be submitted or status is not tracked.	Provides a structured workflow for obtaining and monitoring co-author approvals.
FR-005	Functional	The system shall provide the Research Dean with fund burn-up analytics and research/publication performance information.	Medium	Pass: The Research Dean can view grant fund utilization and relevant publication metrics. Fail: Analytics are	Helps the Research Dean monitor sponsored-fund utilization and faculty

				unavailable, incomplete, or inconsistent with stored records.	research performance.
NFR-001	Non-Functional	The system shall maintain an immutable, tamper-evident audit ledger for financial approvals and journal/publication status modifications.	High	Pass: Audit records cannot be silently altered or deleted and changes to financial approvals and publication status are traceable. Fail: A user can modify or remove audit history without detection.	Protects the integrity and accountability of financial and publication workflow records.
NFR-002	Non-Functional	The system shall provide responsive access to grant, expense, publication, approval, and analytics operations under expected peak usage.	High	Pass: Benchmarking under simulated peak load demonstrates acceptable response performance without service degradation. Fail: Response times become unacceptable or operations fail under expected load.	Ensures the portal remains usable when multiple faculty researchers and administrators access it concurrently.

2. UML Use-Case Diagram:

Actors: Faculty Researcher and Research Dean.



3. Use Cases:

UC-01 — Manage Grant

Primary Actor: Faculty Researcher

Purpose: Create, view, and maintain research grant details including allocation and approved funding limits.

Related Requirement(s): FR-001

UC-02 — Record Research Expense

Primary Actor: Faculty Researcher

Purpose: Record equipment procurement and other research expenses against an active grant.

Related Requirement(s): FR-002

UC-03 — Update Remaining Budget

Primary Actor: Included behavior

Purpose: Calculate the remaining grant balance after a valid expense is recorded.

Related Requirement(s): FR-002

UC-04 — Manage Publication

Primary Actor: Faculty Researcher

Purpose: Record and maintain publication information, including indexing and citation details.

Related Requirement(s): FR-003

UC-05 — Request Co-author Approval

Primary Actor: Faculty Researcher

Purpose: Submit a publication for co-author approval and track its approval status.

Related Requirement(s): FR-004

UC-06 — Track Publication Metrics

Primary Actor: Faculty Researcher / Research Dean

Purpose: Maintain and view publication metrics such as indexing status and citations.

Related Requirement(s): FR-003, FR-005

UC-07 — View Fund Analytics

Primary Actor: Research Dean

Purpose: View grant fund utilization/burn-up information and remaining funding.

Related Requirement(s): FR-005

UC-08 — View Research Metrics

Primary Actor: Research Dean

Purpose: View research and publication performance information.

Related Requirement(s): FR-005

UC-09 — Review Approval

Primary Actor: Research Dean

Purpose: Review and monitor publication/co-author approval status.

Related Requirement(s): FR-004

4. Core Use-Case Flow — Record Research Expense

Primary Actor

Faculty Researcher

Preconditions

The researcher is authenticated; an active research grant exists; the researcher has permission to record expenses.

Postconditions

On success, the expense is recorded, total expenditure is updated, and the remaining grant balance is recalculated and displayed. On failure, the invalid expense is not recorded and the balance remains unchanged.

Main Success Scenario

1. Researcher logs into the system.
2. System displays available grants.
3. Researcher selects an active research grant.
4. Researcher selects “Record Research Expense.”
5. System displays the expense-entry form.
6. Researcher enters expense details and amount.
7. Researcher submits the expense.
8. System validates the expense information.
9. System checks the available grant balance.
10. System confirms the expense is within the approved grant limit.
11. System records the expense.
12. System updates expenditure and recalculates the remaining balance.
13. System displays confirmation and the updated balance.

Alternate Flow — Expense Exceeds Grant Limit

1. Researcher submits an expense.
2. System validates the amount.

3. System compares the amount with the remaining grant balance.
4. The amount exceeds the available balance.
5. System rejects the expense and does not record it.
6. System informs the researcher that the expense exceeds the grant limit.
7. Researcher modifies the amount and may resubmit.